

Title

Translating Pain Management Knowledge Into Home-Based Ambulatory Care: Implementation Outcomes From a Cluster Randomized Trial

Abstract

Community-based older adults often experience inadequately managed chronic pain due to organizational and provider-level barriers. We conducted a pragmatic trial in 22 ambulatory nursing services comparing individualized provider support, digital education, and control conditions (total 190 baseline, 144 follow-up due to 24% attrition). The individualized provider intervention improved documentation of pain assessments compared to controls; however, this improvement in process measures did not translate to meaningful changes in medication appropriateness or patient-reported pain severity at follow-up. The digital education platform was adopted inconsistently by participating providers and showed similarly limited impact on primary clinical endpoints. Qualitative interviews revealed that substantial structural barriers--including insufficient staffing levels, lack of financial reimbursement mechanisms for pain management activities, and competing clinical demands--constrained implementation of any intervention across all study arms. These pragmatic findings underscore that isolated provider- or system-level interventions may be insufficient without concurrent policy reforms that enable dedicated time and resources for comprehensive pain management in ambulatory elderly populations.